Winston Zhang

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EXPERIENCE

Sydney Motorsports Formula Student Team — Sydney, Australia
Aerodynamics Engineer — *2022.07 - 2023.12*
Aerodynamics Section Head — *2023.12 - Current*

Dovetail Electric Aviation — Sydney, Australia
Computational Fluid Dynamics Intern — *2023.09 - 2024.01*

TECHNICAL SKILLS

Composite Manufacturing - Prepreg and Hand Layup

Contributed to the manufacturing of an aerodynamics package for Sydney Motorsports using both wet layup and prepreg techniques. Designed and manufactured various molds for hand layup processes through a combination of hand sculpting, CNC hot wire cutting, and 3D printing.

Computer Aided Design - SOLIDWORKS and Autodesk Inventor

Participated in the design team for the aerodynamic package of Sydney Motorsports and created baseline simulation geometry for Dovetail Electric Aviation. All CAD projects were fully parameterized to enable simple design iterations. Designed multiple iterations of a custom mouse in Autodesk Inventor and modified various online models in Fusion 360 to suit specific needs.

Numerical Simulation - SOLIDWORKS and ANSYS

Developed a semi-automated workflow to test aerodynamic component designs over a range of angles for sensitivity studies with Sydney Motorsports. Created a CFD model of an aircraft engine assembly, including a mesh analysis study and radiator simulation to investigate cooling effects for Dovetail Electric Aviation. Conducted thermal studies in ANSYS Fluent and Steady State Thermal to analyze human comfort and airflow. Performed various FEA and topology optimization studies in SOLIDWORKS and ANSYS Steady State Structural for Sydney Motorsports.

Programming - MATLAB and Python

Proficient in writing analysis code in MATLAB and Python. Conducted a blade element analysis in MATLAB to evaluate propeller performance under different flight conditions. Developed custom FEM code and validated it against a commercial Steady State Thermal package.

TECHNICAL PROJECTS

Semi-Automated CFD Workflow

Enhanced the robustness of the semi-automated CFD workflow used by Sydney Motorsports by experimenting with mesh settings, enabling more designs to be iterated through the workflow.

Aerodynamic Package for FSAE Car

Planned the manufacturing, mounting, and validation of Sydney Motorsports' second in-house aerodynamics package. Utilized a variety of manufacturing methods, including FDM printing, CNC hot wire cutting, laser cutting, and hand sculpting, for mold production. Components were fabricated using conventional wet layup processes under vacuum with a combination of bi-axial and uniaxial carbon fiber. Improved mounting solutions to streamline track-side assembly and maintenance. Optimized the rib and spar system design for the front wing by evaluating various materials to balance performance, cost, manufacturing, and time constraints.

Custom Cloud Storage Server

Built a custom server using e-waste and second-hand parts for self-hosted cloud storage solutions similar to OneDrive, configured with RAIDZ5 for redundancy. The system also functions as a NAS for additional storage and offers remote access through VPN.

Robotic Arm

Designed a 6-axis robotic arm out of NEMA17 and NEMA11 motors with a target payload of 1.5 kilograms. The hardware consists of two cycloidal gearboxes with 1:50 and 1:20 reduction ratios, as well as a turret-style planetary gearbox.

Custom Computer Peripherals

Designed a computer mouse that incorporates anthropometric sizing considerations and tested prototypes, resulting in a final weight of 32 grams. This included a fully custom shell and hollow base for weight reduction. Created custom cases for the Lily 58 split ortholinear keyboard. Hand-soldered hot-swappable sockets and SMD diodes to the PCB and through-hole soldered the Pro-Micro style microcontroller.

TRANSFERABLE SKILLS

Leadership and Management

Leading a team of 10 student engineers in designing and validating the next-generation aerodynamics package. Delegated tasks to team members based on their strengths and experience, ensuring new engineers are sufficiently upskilled. Managed the production timeline for the current aerodynamics package.

Problem Solving

Effectively addressed design challenges by modifying parts to improve manufacturability without compromising functionality. Responded quickly to manufacturing issues under tight deadlines.

Communication

Facilitated clear communication of technical details between departments to ensure the feasibility and integration of designs across the team.

Time Management

Maintained personal design deadlines for the Formula SAE team and Dovetail Electric Aviation while studying full-time. Ensured production timelines and goals were met without overburdening team members or compromising quality.

EDUCATION

THE UNIVERSITY OF SYDNEY Expected Graduation: 2025
Bachelor of Mechanical Engineering Honours (Fluid Mechanics) - ~~*Credit WAM*~~ *2021 - Current*